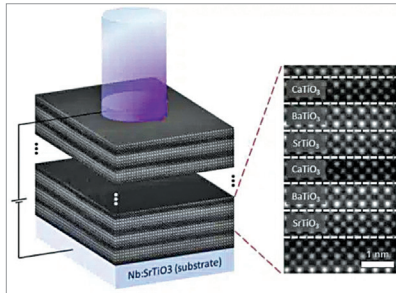


Metamaterial enhances ultrasound imaging and non-destructive testing

A research team at KIMM's Department of Nano Devices & Displays and SNU's Department of Mechanical Engineering has created a metamaterial capable of converting linear ultrasonic wave vibrations into circular vibrations, featuring a three-dimensional microstructure. They discovered the anisotropic properties of an ultrasonic medium enabling this conversion and designed a metamaterial with three-dimensional cylindrical holes to achieve it. This innovation allows for the efficient generation of circular shear ultrasonic waves using simplified cylindrical microstructures and low-loss metal materials. They have developed a new ultrasonic mode that can enhance the defect detection capability of existing ultrasonic technologies and apply this technology in industrial ultrasonic non-destructive testing and ultrasound imaging.

New technique enhances solar cell efficiency by 1000 times

At Martin Luther University Halle-Wittenberg (MLU), researchers developed a technique to enhance solar cell efficiency by 1000 times. They created crystalline layers of barium titanate, strontium titanate, and calcium titanate



For their new approach, the researchers combined three crystal materials (Credit: Uni Halle/Yeseul Yun)

stacked in a lattice formation. Exposed to laser light during measurements, the new material showed a current flow increase of up to 1000 times compared to pure barium titanate of similar thickness, despite reducing the proportion of

barium titanate by nearly two-thirds. This could lead to more efficient and cost-effective solar panels, requiring less space and benefiting urban areas. The team's work could significantly boost solar energy adoption, reducing reliance on fossil fuels and combating climate change.

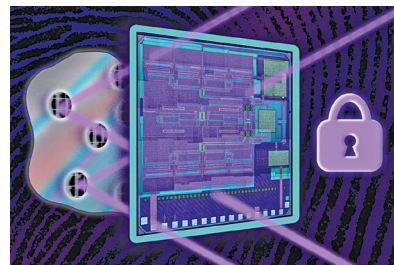
Flexible, wireless and unbreakable optical sensors

Researchers from the Osaka University have developed a flexible, wireless optical sensor using carbon nanotubes and organic transistors on a thin polymer film. Unlike traditional rigid optical sensors, this innovative sensor remains

functional even when bent or crumpled. It detects both visible and infrared light by heating carbon nanotubes, which then generate a voltage signal. The sensor's sensitivity is enhanced by doping the nanotubes with chemical carriers. Integrated with wireless technology, organic transistors on the polymer substrate convert the voltage signals into an image signal, transmitted wirelessly to a computer via Bluetooth. The prototype has successfully sensed heat from various objects and the flow of glucose through tubes. With its high sensitivity, durability, and wireless capabilities, this sensor has potential applications in non-destructive imaging, wearable devices, and soft robotics, offering a simpler way to evaluate liquid quality without sampling.

Inexpensive tag for identification and security applications

MIT researchers have created a small security tag using terahertz waves, which are quicker and smaller than radio waves, offering better protection than traditional radio-frequency identification (RFID) tags. To prevent counterfeiting,



A cryptographic tag developed at MIT uses terahertz waves to authenticate items by recognising the unique pattern of microscopic metal particles that are mixed into the glue that sticks the tag to the item's surface (Credits: Image: Jose-Luis Olivares, MIT; Chip courtesy of the researchers)

they embedded microscopic metal particles in the adhesive, forming a unique pattern that acts like a fingerprint when scanned with terahertz waves. A machine-learning model accurately detects tampering by comparing over 99% accuracy of glue patterns. This

inexpensive tag can be used in large supply chains and on small items, such as medical devices. Despite challenges like high transmission loss and a limited range of terahertz waves, the team sees potential for their use in identification and security applications.

Wearable technology for real-time emotion recognition

Researchers at UNIST's Department of Material Science and Engineering have developed a real-time human emotion recognition technology, poised to revolutionise industries, particularly in wearable systems. Their multi-modal system combines verbal and non-verbal data for accurate emotion analysis. The core technology is the personalised skin-integrated facial interface (PSiFI), a self-powered, flexible, and